

Vision-Only Latent Diffusion Super-Resolution without T2I Pretraining

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Completed-experiment snapshot through Stage 2 XL candidate selection

Abstract

This report summarizes a work-in-progress x4 super-resolution system trained directly in vision space, without importing a pretrained text-to-image diffusion backbone. The completed pipeline consists of a factor-4 VAE, deterministic LR-to-HR-latent pretraining, conditional latent diffusion, and condition-start fine-tuning on a 103k-image photo split. The latest completed XL-stage result is a 500M-class condition encoder trained with a stronger degradation curriculum. Its Stage 2 candidates were compared on the same fixed validation images, and the 72k-step checkpoint was selected for the next diffusion stage.

Objective

The target task is single-image x4 restoration: 128x128 LR to 512x512 HR, with a later 192x192 to 768x768 extension. The design goal is a practical latent diffusion SR model trained from project checkpoints only. The system intentionally avoids Stable Diffusion, T2I backbones, text conditioning, and imported generative priors from pretrained diffusion models.

Architecture

The model is split into a factor-4 VAE, an LR condition encoder, and a conditional diffusion U-Net. HR images are encoded into 128px, 16-channel latents. Degraded LR images are encoded into condition latents. The U-Net denoises HR latents using the condition latent, timestep embedding, and domain ID. The VAE decoder maps the denoised latent back to a 512px HR image.

Training Stages

- 1 Stage 1 trains the factor-4 VAE on HR crops.
- 2 Stage 2 trains a deterministic condition encoder to predict HR latents from degraded LR inputs.
- 3 Stage 3 trains conditional latent diffusion in latent space.
- 4 Stage 4 fine-tunes diffusion from the condition latent at low timesteps.
- 5 XL variants widen and deepen models while using partial initialization where tensor shapes match.

Completed Implementation

The repository includes manifest-based dataset loading with explicit domain IDs, on-the-fly x4 degradation, AutoencoderKL training, deterministic latent pretraining, conditional latent diffusion, DDIM/img2img validation sampling, condition-start fine-tuning, bf16 CUDA/ROCm training paths, W&B; logging, fixed validation samples, checkpoint evaluation, VM recovery scripts, and partial checkpoint initialization for widened/deepened models.

Data and Degradation

Item	Value
Task	x4 SR, 128px LR to 512px HR
Active large-scale domain	Photo
Train split	103,450 images
Validation split	100 fixed images
VAE latent	128px, 16 channels

Primary dtype	bf16
Latest completed XL degradation	photo_v3_noise_mix

Completed Results: Stage 1 VAE

The selected Stage 1 VAE checkpoint is autoencoder_photo10k_b16_eval_online/checkpoints/best_eval_recon.pt. It was selected at the 50k-step checkpoint.

Metric	Value
eval/recon	0.01198
eval/kl	9.38684
eval/psnr	40.19

Completed Results: Photo100k Baseline

Stage	Run	Finished step	Selected result
Stage 2	latent_pretrain_photo100k_b64	30k	best latent loss 0.21230
Stage 3	diffusion_photo100k_b32	60k	decoded PSNR 24.56 diagnostic
Stage 4	diffusion_photo100k_b32_stage4_condition	5k	sampled val100 PSNR 25.4072

On sampled validation, Stage 4 condition-start slightly improved the Stage 3 checkpoint: $25.4072 - 25.3745 = +0.0327$ PSNR.

Completed Results: Photo100k v2

Comparison	Sampled val100 PSNR
Bicubic baseline	22.4103
Stage 3 v2 SR	22.6699
Stage 4 v2 SR	22.8426

Stage 4 v2 improved over Stage 3 v2 by +0.1727 PSNR and beat bicubic by +0.4323 PSNR, with 81 wins and 19 losses in per-image comparison. Remaining artifacts included occasional color/contrast overshoot and small cyan/green sampling artifacts.

Completed Results: Stage 2 XL Condition Encoder

A smaller photo_v3_noise_mix Stage 2 run was stopped at step 12,700 after validation latent loss plateaued near 0.282. The XL condition encoder then completed 80k steps with a larger 500M-class full-path target.

Checkpoint	Step	eval/latent_loss	eval/decoded_psnr
Best latent	66k	0.27230	21.38
Best PSNR proxy	72k	0.27940	21.52
Final/latest	80k	0.27592	21.51

The three Stage 2 XL candidates were compared on the same 100 validation images by decoding condition-only outputs through the selected VAE. The 72k-step checkpoint had the best decoded condition-only PSNR and the lowest decoded MSE, so it was selected as the condition encoder for the next XL diffusion experiment.

Candidate	Mean decoded MSE	Mean decoded PSNR
Best latent, step 66k	0.029092	22.8017
Latest, step 80k	0.028277	22.8901

Best PSNR proxy, step 72k	0.028161	22.9029
Bicubic reference	--	22.3599

Current Scope

This document records completed experiments only. XL diffusion training and any metrics from runs still in progress are intentionally excluded until those runs finish and their checkpoints can be compared on fixed validation samples.

Limitations

This is an engineering report for an active experiment, not a claim of a final published method. The large-scale completed results are photo-domain only, while anime/illustration domain conditioning remains implemented but not yet validated at the same scale. The reported sampled PSNR values are useful for internal checkpoint comparison, but they are not a full benchmark suite against external SR methods.

References

Rombach et al., High-Resolution Image Synthesis with Latent Diffusion Models, CVPR 2022.

Saharia et al., Image Super-Resolution via Iterative Refinement, IEEE TPAMI 2022.